

Software Requirements Specification (SRS)

Project: AI Test Case Generator & QA Assistant using RAG

1. Introduction

Purpose, Scope, Definitions, Intended Audience, References, Overview.

2. Overall Description

Product Perspective, Product Functions, User Classes, Operating Environment, Constraints, Assumptions.

3. System Features

Authentication, Project Management, Document Management, RAG Processing, AI Chat, AI Test Case Generation, Test Case Repository, Export Module, User Profile.

4. External Interface Requirements

UI, Software, Hardware and Communication Interfaces.

5. Non-Functional Requirements

Performance, Security, Reliability, Scalability, Availability, Usability, Maintainability, Compatibility.

6. Database Design

ER Diagram, Tables, Relationships, Data Dictionary.

7. Use Cases

Register, Login, Create Project, Upload Document, Generate Test Cases, AI Chat, Export, Delete Project.

8. AI & RAG Workflow

Upload, Extraction, Chunking, Embeddings, Vector Storage, Retrieval, Prompting, Response.

9. Technology Stack

Angular 21, .NET 10, MySQL 8, EF Core 10, JWT, OpenAI, ChromaDB, PdfPig, OpenXML, ClosedXML, QuestPDF, Swagger.

10. Future Enhancements

Jira/Azure DevOps integration, Coverage Matrix, Test Script Generation, Collaboration.